***NOTE***

Data assimilation is like giving your model a regular dose of reality. It compares what the model thinks should happen with what actually happened, then adjusts accordingly.

This makes the model not just a static representation of the system that predicts outcomes based on sensor data, but a living, evolving digital counterpart.

### *REAL WORLD EXAMPLE: PREDICTING THE WEATHER*

Imagine you are planning to go on a hike at the weekend. You will probably look at your favorite weather app to see whether rain is in the forecast for the day you plan to go. Beyond the simple icon of clouds and raindrops lies a sophisticated chain of models, simulations, and real-world data—a comprehensive digital twin of the global weather maintained by institutions like the European Center for Medium-Range Weather Forecasts (ECMWF). The ECMWF maintains multiple numerical models of the atmosphere, known as general circulation models, that it combines with recent sensor observations from global weather stations and remote satellite data (using data assimilation) to get the best possible estimate of the current state of the atmosphere. The updated model becomes the foundation for predicting upcoming weather patterns and whether rain is likely.

The ECMWF is currently implementing an earth-scale digital twin called Destination Earth (https://destine.ecmwf.int/) that will provide a planetary-scale digital replica based on observations and simulations. The first two capabilities include multi-decade simulations of climate change, and on-demand simulations of weather-induced extremes, with the intent being to more accurately predict environmental disasters and mitigate resultant crises. Example output from a simulation run within the climate change adaptation digital twin that is used to inform wind farm design is shown in figure 1.10.